

AI-Native CRM — 4-Week Build Roadmap

What we're building. A configurable, AI-native CRM platform that captures leads across every channel, qualifies and scores them automatically, and drives them to close — with a human always in the loop. The platform is **domain-agnostic**: entities, pipeline stages, scoring signals, channels, and copy are **configuration, not code**. This pilot is configured for **real-estate sales**, but the same engine runs any vertical CRM.

The plan. 4 people · 4 weeks · a live pilot at the end of Week 4. The product's front-end is already built; these four weeks add the real backend, AI, integrations, data layer, and hardening for launch.

1. Team

Role		Focus
2 ×	AI Engineers	AI gateway, conversational agents, scoring & matching, copilot, analytics intelligence, evaluations
2 ×	Full-stack Engineers	Backend & data platform, channel integrations, app wiring, security, deployment

2. Technology approach (high level)

AI models — closed (proprietary) LLMs behind a model-agnostic gateway. We standardise on best-in-class **hosted, closed frontier models**, accessed through a gateway that routes each task to the right model tier and lets us **swap providers without code changes**. Every AI call is traced, cost-capped, and guardrailed (answers are grounded to real data — no fabricated facts or prices).

Model tier	Used for
High-volume tier (fast, low-cost closed LLM)	Conversational qualifying, classification, extraction, scoring
Reasoning tier (frontier closed LLM)	Executive brief, root-cause analysis, natural-language → workflow & analytics
Embeddings (closed model) + vector store	Matching, de-duplication, semantic search, retrieval
Speech (closed STT/TTS)	Voice calls and meeting transcription

Platform (managed cloud). Web app, API services, a relational database + vector store, cache, a background-job/queue layer for autonomy & automations, realtime updates, object storage, authentication with role-based access, and full observability.

Channels (pluggable connectors). WhatsApp · SMS · Email · Voice · Web/portal lead sources — each a configurable connector, monitored for health.

3. Timeline (4 weeks)

3a. Database design & validation track

The data layer is **designed and validated in Week 1**, then evolves additively as features land — no internal schema detail here, just the phases and their done-criteria.

Phase	When	Outcome	Validated when
Model design	Week 1 · Days 1–2	Entity-relationship model — a generic CRM core plus the real-estate configuration	Model reviewed & signed off
Build	Week 1 · Days 2–4	Schema, migrations, and seed data	Live on the staging environment
Validate	Week 1 · Day 5	Integrity & constraints, access-scoping by role, sample-data load, backup/restore	All data + access checks pass ✓
Evolve	Weeks 2–4	Additive, zero-downtime migrations as new features need them	Each change reviewed & reversible

4. Week-by-week feature plan

Every feature is listed under the week it ships. Sequence is dependency-driven — nothing reasons over data that isn't captured yet, and nothing aggregates before the data exists. ★ = headline feature · ◇ = stretch (cut first if behind).

Week 1 — Foundations & data

Outcome: the app runs on real data; the data model is designed & validated.

- Platform & delivery — cloud scaffold (web app, API services), CI/CD, environments, deployment
- Authentication + **role-based access** (Manager · Sales Agent · Telecaller) + tenant scoping
- Database: design → build → validate** — see §3a
- AI gateway** — closed LLMs behind a model-agnostic router, with tracing, cost caps, grounding guardrails; evaluation harness
- ★ **Lead scoring engine + explainability** (across the configured signal categories) + adjustable signal weights
- ★ **Inventory/catalog matching** (embeddings + vector store)
- Realtime-updates layer
- Wire core screens to **live data, no mock data** — role-based **home dashboards**, worklist, pipeline, inventory/catalog

Week 2 — Capture & the AI agent

Outcome: the AI handles conversations end-to-end, on its own.

- **Omni-channel capture** — WhatsApp, Email, SMS, Web/portal lead sources (Voice/IVR ◇)
- ★ **Conversational qualifying agent** — multilingual; qualifies, scores, matches the right offering, and books the next step
- ★ **AI Inbox (live)** — omni-channel threads, status funnel, metrics, **real-time capture sidebar**
- **Human-in-the-loop takeover** of any conversation
- ★ **Natural-language command bar** — speak/type a sentence → structured record; plus global search & navigate
- **De-duplication + merge**
- **Enrichment / sourced intel** + eligibility checks
- **Live scoring from message signals** — re-score + re-rank on every new message
- **Overnight auto-capture + Morning Brief**

Week 3 — Agent workflow

Outcome: the full daily workflow works with AI assist.

- **Visual pipeline** — configurable stages, drag-and-drop, required remarks, stage automations, quote/invoice draft
- **AI next-step suggestions** (accept the recommended move)
- **Inventory/catalog management** — live availability (configurable entities)
- **AI-ranked worklist** — "why ranked", SLA/overdue, re-ranks live
- ★ **Unified customer 360** — one timeline across every channel; send-from-profile; explainable score view
- ★ **Agent copilot** — grounded in the rep's own book; answers *and* acts (create, book, search, root-cause)
- **AI-drafted next-best messages** (per stage)
- **Approvals (human-in-the-loop)** — all approval types (outbound, new-lead, sequence, stage-move, field-update, duplicate-merge)
- **Meetings** — transcription + sentiment + summary → tasks
- **Task management**
- **Team performance management** (for managers)
- Logistics / fulfilment movement ◇

Week 4 — Intelligence, automation & launch

Outcome: intelligence, automation, hardening, and go-live.

- ★ **Analytics suite** — full dashboards (funnel, velocity, win/loss, source ROI, leaderboard, capture mix, score-validation, automation health, revenue leakage)
- ★ **Executive brief** — auto-generated, week-over-week
- **Natural-language analytics queries** (guarded)
- **Lost-reason analysis + risk & opportunity signals**
- **AI sales playbook** (what wins deals)
- ★ **No-code automations** — natural-language → workflow, with run history
- **Broadcast** — audience segments + scheduling, human-approved

- **Source onboarding wizard** + connector health monitoring
- Usage & billing + **audit trail**
- **Hardening** — security, rate limits, PII handling, load test
- **AI evaluations & quality** — grounding (no fabricated facts)
- **UAT → production launch → monitoring + runbook**

5. Configurable for any CRM — not just real estate

The platform is vertical-agnostic. Standing up a new domain is **configuration, not a rebuild** — the same AI engine, data core, channels, and workflow serve any business. What changes per domain:

Configuration	Real-estate example	Generic
Entities & fields	Buyers, projects, units	Contacts, products, accounts
Pipeline stages	Enquiry → site visit → booking → registration	Any defined sales/service stages
Scoring signals & weights	Budget fit, locality, visit intent, loan readiness	Any intent/fit signals
Channels & sources	Property portals + WhatsApp	Any lead sources & channels
Copy, prompts & playbooks	Real-estate tone & rules	Domain tone & rules
Roles	Manager / Sales Agent / Telecaller	Any role hierarchy

Real estate is simply the launch configuration; the same product serves sales, services, education, healthcare, and more.